

# MOHAMMED AFFAAN KHAN

Agentic AI Engineer · GenAI & LLM Systems · Conversational Voice AI · RAG & Vector Search · AI Agent Development

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3 YRS 8 MOS TOTAL · 3 YRS 1 MO IN AGENTIC AI, GENAI & CONVERSATIONAL VOICE AI · 14+ SYSTEMS SHIPPED

<500 ms voice latency · 10,000+ concurrent calls · 99.9% uptime SLA · 85% faster clinical reporting · 45% lower operating cost

## PROFESSIONAL SUMMARY

Agentic AI Engineer with 3+ years across agentic AI, generative AI, conversational voice AI, and machine learning, and 14+ systems taken from architecture to live deployment — multi-agent orchestration, RAG pipelines, real-time conversational voice AI, and MCP-based tool integration. Owns the full lifecycle from AI architecture and prompt engineering through evaluation, microservices deployment, and MLOps across AWS Bedrock, Azure OpenAI, and Google Vertex AI. Shipped voice agents at sub-500 ms STT→LLM→TTS latency across 10,000+ concurrent calls at 99.9% uptime, cut clinical reporting time 85%, and reduced client operational cost 45% through autonomous workflow automation. Full-stack and distributed-systems foundation (Python, FastAPI, Node.js, React, PostgreSQL, Kubernetes) with an AI-native development workflow built on Claude Code, Codex, and Cursor.

## TECHNICAL SKILLS

**Agentic AI & Orchestration** AI Agent Development, Multi-Agent Systems, LangChain, LangGraph, CrewAI, AutoGen, Model Context Protocol (MCP), MCP Server Development, OpenClaw, Tool / Function Calling, ReAct & Planner-Executor Patterns, Human-in-the-Loop Approval, Agent Memory & State, Guardrails

**LLMs & Prompt Engineering** GPT-4o, Claude (Sonnet / Opus), Llama 3, Gemini, Prompt Engineering, Context Engineering, Structured Outputs, Chain-of-Thought Design, Fine-Tuning, RLHF, LLM Routing & Fallback, Token & Cost Governance

**RAG & Vector Databases** Retrieval-Augmented Generation, Pinecone, pgvector, AlloyDB (ScaNN), FAISS, Weaviate, Qdrant, LlamaIndex, Hybrid Search, Re-Ranking, Chunking Strategies, BGE Embeddings, Semantic Search, Self-Updating Knowledge Pipelines

**Conversational & Voice AI** Millis.ai, ElevenLabs, Whisper STT, Google TTS, WebRTC, SIP / VoIP, Real-Time Streaming, Barge-In & Turn-Taking, Voice Cloning, Emotion-Adaptive TTS, Intent Classification, NLU

**Cloud, MLOps & Evaluation** AWS (Bedrock, Lambda, EC2, S3), Microsoft Azure (Azure OpenAI Service, Azure AD / Entra ID, AKS), Google Cloud (Vertex AI, BigQuery, AlloyDB), NVIDIA NIM / NVIDIA LLM APIs, OpenRouter, Kubernetes, Docker, Harness CI/CD, GitHub Actions, Model Registry, LLM Tracing & Observability, Evaluation Harnesses, A/B Testing, GPU Optimisation, INT8 Quantization

**AI Tooling & Automation** Claude Code, OpenAI Codex, Cursor, GitHub Copilot, Microsoft 365 Copilot, n8n, Apify, Zapier, Make

**Backend & Frontend** Python, FastAPI, Node.js, Express, Microservices, Event-Driven Architecture, REST APIs, WebSockets / Socket.IO, PostgreSQL, Redis, BullMQ, Supabase, Alembic, Sequelize, Multi-Tenant SaaS, RBAC, Stripe, React, Vite, TypeScript, Tailwind CSS, i18next (RTL / LTR)

**ML & Computer Vision** PyTorch, TensorFlow, HuggingFace Transformers, Scikit-learn, XGBoost, U-Net, YOLOv5, GRU, GANs, OpenCV, VTK, SimpleITK, DICOM, 3D / MPR Rendering

## PROFESSIONAL EXPERIENCE

### Maskan Technologies

May 2025 – Present · 1 yr 4 mos

AI Developer — Agentic AI & Generative AI

Bangalore, India

End-to-end **ownership** of AI **solution design** through production delivery as the firm's primary AI engineer — **system design**, technology selection, and **scalability** and **security** trade-offs for **enterprise** agentic systems, conversational voice AI, RAG platforms, and ML services across MedTech, FinTech, MarTech, and hospitality clients in India, the UK, and the GCC. Run **proof of concept** through to production, define **reference architecture** and engineering **best practices**, and drive **cross-functional** delivery with client stakeholders.

**Autonomous GTM Agent Platform ("PipelineIQ") — Agentic Lead Generation & Qualification** | LangGraph · MCP · Claude · Apify · Meta Lead Ads API · n8n · pgvector · AWS Bedrock

- Architected a **multi-agent GTM system** that runs the entire top of funnel autonomously: a research agent mines market topics and keywords, a content agent drafts and publishes **LinkedIn** posts on schedule, a prospecting agent discovers and enriches leads, and a qualification agent scores them against ICP criteria — coordinated by a **LangGraph** supervisor with shared state, retries, and human-in-the-loop approval gates.
- Exposed every capability — scraping, enrichment, CRM writes, publishing, ad-platform reads — as **Model Context Protocol (MCP)** tool servers, enabling zero-code capability addition and provider swapping across Claude, GPT-4o, and **AWS Bedrock**-hosted models.
- Wired multi-source capture through **Apify** actors and **Meta Lead Ads** webhooks into the MaskanX CRM, where **autonomous voice agents dial each new lead, qualify it in conversation, and advance the lead stage** with zero human dispatch — closing the loop from campaign launch to sales-ready pipeline.
- Built governance into the agent loop: per-run token and cost budgets, rate-limit-aware scheduling, prompt-injection guarding on untrusted scraped content, and full run tracing for replay, evaluation, and prompt regression testing.

**Conversational AI Voice Agent Platform — Real-Time Voice Intelligence** | Millis.ai · Whisper STT · GPT-4o · LangChain · ElevenLabs · Kubernetes · Azure OpenAI · n8n

- Architected an end-to-end real-time voice platform sustaining **10,000+ concurrent calls** at **99.9% uptime SLA** and **<500 ms end-to-end STT→LLM→TTS latency** in production.
- Engineered the **agentic decision loop** in LangChain with persistent multi-turn memory, dynamic role-switching, and function calling into CRM and booking **microservices** — automating **2,000+ daily interactions** and cutting operational cost **45%**; orchestrated post-call **n8n** workflows for CRM enrichment, scheduling, and escalation routing.

- Deployed on **Kubernetes** with autoscaling absorbing 10× traffic spikes and production **monitoring**, alerting, and **incident** response holding **reliability** at the contracted SLA; cut inference spend **35%** via INT8 quantization, batched GPU utilisation, and cost-aware **multi-cloud** model routing across **Azure OpenAI** and **AWS Bedrock**.

**Real-Time AI Avatar with Retrieval-Augmented Generation — Multimodal Digital Human** | *Pinecone · LlamaIndex · GPT-4o · ElevenLabs · Facial Animation Engine*

- Built a multimodal digital human delivering **200–400 ms response latency** through a synchronised **RAG** pipeline, neural TTS, and a real-time facial animation engine, deployed to live production audiences.
- Indexed **200+ hours** of structured content into 768-dimension **Pinecone** embeddings with hybrid retrieval and re-ranking; built a self-updating knowledge pipeline that re-indexes automatically on source change.

**NHS AI-Powered Clinical Workflow System — Lung Nodule Detection & Reporting** | *PyTorch · U-Net · VTK · FastAPI · PostgreSQL · React · DICOM · Azure AD*

- Engineered a deep-learning CT analysis pipeline using **U-Net** segmentation (**Dice > 0.85**) and a hybrid malignancy classifier at **92% accuracy** for lung-nodule risk stratification.
- Delivered a full role-based clinical platform (Pulmonologist / Radiologist / Admin) with **Azure AD (Entra ID)** SSO and RBAC, designed to NHS **security** and **compliance** requirements and documented as a **reference architecture** with **data flow** diagrams — covering referral management, nodule tracking, MDT discussions, and clinical decision logging over REST **microservices**.
- Built a 2D/3D CT visualisation suite (VTK MPR / Glassy viewer) with auto-generated PDF/MP4 reports — cutting radiologist reporting from **30 minutes to ~5 minutes (85% reduction)**, with pilot data showing a **40% workflow efficiency gain**.

**Xplortiv — AI Recommendation & Discovery Engine** | *PyTorch · GRU · BGE Embeddings · pgvector / AlloyDB · BigQuery · Vertex AI*

- Architected a booking-conversion-optimised ranking system: a GRU short/medium/long-term event encoder with task-conditioned gating across **9 prediction heads** (skip, watch, book, verify, capture) under prioritised loss weights.
- Designed a natural-language venue **semantic search** engine (“date night in Glasgow”, “rooftop bar nearby”) on BGE-small-en 128-D embeddings served via **pgvector / ScaNN** on **AlloyDB** at **sub-50 ms** retrieval; built the **MLOps** pipeline on **BigQuery** offline assembly, **Vertex AI** training and model registry, and INT8 quantization for production inference.

**MaskanX — Multi-Tenant AI Marketing & CRM SaaS** | *MERN · Claude API · Redis · BullMQ · Temporal · Agentic Email & Voice Automation*

- Built the agentic layer unifying lead management, campaign management, AI content generation, and autonomous email/voice outreach behind one multi-tenant control plane, with **Redis + BullMQ** job orchestration and Temporal-backed long-running workflows.
- Implemented the AI lead-qualification service that ingests leads from ad platforms and scrapers, triggers autonomous voice qualification, and writes stage transitions back to the CRM in real time.

**Maskanza — Multi-Tenant Restaurant Management SaaS** | *Node.js · Express · PostgreSQL · React · Socket.IO · Stripe · i18next*

- Shipped a production multi-tenant SaaS spanning POS, Kitchen Display System, QR ordering, inventory, reservations, and analytics across **132 API endpoints** and **53 frontend pages** — with real-time order orchestration over Socket.IO, Stripe subscription billing with webhook lifecycle management, a conversational AI booking agent, **9-language i18n** including RTL Arabic, and **23 RBAC permission scopes**.

**OpenMontage, SiteChat & Estimate256 — Agentic Video, RAG Chat & ML Estimation** | *LangChain · Remotion · FAISS · OpenRouter · FastAPI · XGBoost · Alembic*

- **OpenMontage**: instruction-driven agentic video pipeline with 11 specialised pipeline types orchestrated by an LLM agent via declarative YAML manifests, plus a dynamic tool registry giving zero-code provider swapping and cost governance across multi-provider asset generation.
- **SiteChat**: RAG chatbot platform (URL discovery → FAISS indexing → GPT-4o chat) shipped as a FastAPI + React SPA for zero-code bot creation. **Estimate256**: ML estimation engine with three domain-specific XGBoost models, catalog/billing/auth layers, and Alembic-managed PostgreSQL migrations.

**Independent AI Engineer — Applied AI Development & Specialisation**

**Aug 2023 – Apr 2025 · 1 yr 9 mos**

*LLM Systems · RAG Architecture · Agentic Orchestration · Voice Pipeline Engineering*

*Bangalore, India*

- Built AI systems independently across the full modern stack while transitioning from mechanical engineering into applied AI — LLM integration and prompt engineering, RAG and vector database architecture, multi-agent orchestration (LangChain, LangGraph, CrewAI), and real-time voice pipeline engineering (Whisper, ElevenLabs, Millis.ai).
- Built and published the project portfolio that led directly to the AI engineering role at Maskan Technologies.
- **J.P. Morgan Chase & Co. — Software Engineering Job Simulation** (Forage), completed Feb 2024: configured a local development environment, repaired broken repository files to restore correct web-application output, and integrated **Perspective**, JPMorgan Chase’s open-source visualisation library, to render a live streaming data feed as a real-time trader-facing graph. *Python, TypeScript, React, Git*.
- **Accenture North America — Data Analytics & Visualization Job Simulation** (Forage), completed Feb 2025: cleaned, modelled, and analysed a client dataset, built the supporting visualisations, and packaged the findings into a client-ready deck with a recorded stakeholder readout. *Data modelling, data visualisation, storytelling, strategy, stakeholder communication*.

**BYJU’S**

**Jan 2023 – Jul 2023 · 7 mos**

*Business Development Associate*

*Bangalore, India*

- Engaged **500+ prospective clients**, surfacing product requirements that directly informed EdTech AI feature-roadmap prioritisation.

## EDUCATION

**B.Tech, Mechanical Engineering — Reva University**

**2023**

*AI/ML Electives · Computer Vision · Data Structures & Algorithms · Computational Mechanics*

*Bangalore, India*

- Capstone: **YOLOv5 Infection Detection System** — 95% defect detection accuracy; internally published at Reva University.
- **Certifications**: self-directed specialisation in agentic AI systems, MCP server development, voice pipeline engineering, and vector database architecture. **Languages**: English (Native), Hindi (Native), Urdu (Conversational).